

RICCO CABRAL VENTEREA

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EDUCATION

- PhD** University of California, Riverside *Aug 2025 – Present*
Astronomy
- Università degli Studi di Perugia *Oct 2024 – Jun 2025*
Visiting Fulbright Research Scholar
- BA** Cornell University *Aug 2020 – May 2024*
Astronomy with Astrophysics Concentration *with honors: magna cum laude*
Minor in Physics

RESEARCH EXPERIENCE

Sources and Transients Team, Simons Observatory *Jul 2026 – Present*
Graduate Student Junior Member
Advisors: Drs. John Orlowski-Scherer, Josh Borrow

- Build cross-matching source catalog for data maps and web interface for additional validation.

Department of Physics and Astronomy, University of California, Riverside *Aug 2025 – Present*
Center for Experimental Cosmology and Instrumentation
Graduate Student Researcher
Advisors: Professor Steve Choi, Dr. Junna Sugiyama

- Calculate angular power spectra of the cosmic microwave background from cosmology observatories.

Department of Physics and Geology, University of Perugia, Perugia, Italy *Oct 2024 – Jun 2025*
Fulbright Research Scholar
Advisors: David Pelosi, Professor Nicola Tomassetti

- Created database for solar properties measured by global observatories.
- Developed cross-correlation study between solar modulation and solar proxies.

Cornell Center for Astrophysics and Planetary Science, Ithaca, NY *Jan 2024 – May 2024*
Student Research Assistant III
Advisors: Professor Nicholas Battaglia, Dr. John Orlowski-Scherer

- Developed database containing asteroid thermal emission flux measurements using AWS.

School of Physics and Astronomy, University of Minnesota, Twin Cities, Minneapolis, MN
Jun 2023 – Sep 2024

LIGO Scientific Collaboration
 A3D3 AI Institute

Research Assistant

Advisors: William Benoit, Professor Michael Coughlin

- Work in implementing machine learning algorithms to clean and analyze gravitational wave strain data collected from the Laser Interferometer Gravitational-Wave Observatory Livingston and Hanford sites.
- Generated constant-Q transforms for high signal-to-noise gravitational wave event simulations.
- Developed a user-interface to create distributions of black hole parameter simulations.

Nexus Scholars Program, Cornell University, Ithaca, NY *May 2022 – Jul 2022*

Research Assistant

Advisor: Professor Nicholas Battaglia

- Continued full-time research in measuring thermal emission of objects in the asteroid belt.
- Detected large asteroids with a high degree of confidence and generated light curves. Presented results at a symposium.

Department of Astronomy, Cornell University, Ithaca, NY *Sep 2021 – May 2024*

Research Assistant

Advisors: Professor Nicholas Battaglia, Dr. John Orlowski-Scherer

- Determined flux measurements from thermal emission data of objects in the asteroid belt. Data collected by the Atacama Cosmology Telescope.

National Aeronautics and Space Administration, Remote *May 2021 – Aug 2021*

Lucy Student Pipeline Accelerator and Competency Enabler
 Mission Concept Academy

Astrophysicist

- Submitted a Preliminary Design Review on a science reconnaissance mission for water ice mapping in the lunar South Polar Region.

Summer Student Theoretical Physics Research Session, Remote *Jun 2021*

Student Researcher

Advisors: Professors Jim Gates, Kory Stiffler, Konstantinos Koutrolikos

- Learned topics in supersymmetry with a mathematical emphasis.
- Covered areas in group theory, Lie algebra, differential geometry, tensor algebra, Lagrangian dynamics, Clifford algebra, gauge theory, and supersymmetry algebra.
- Gained experience in LaTeX.

National Aeronautics and Space Administration, Remote *Jan 2021 – Apr 2021*

Lucy Student Pipeline Accelerator and Competency Enabler

Proposal Writing and Evaluation Experience Academy
Technical Team Member

- Submitted a research proposal on human health and performance in low-earth orbit.

Irondale High School, New Brighton, MN

Sep 2019 – May 2020

Student Researcher

Advisor: Shane Wood

- Verified Einstein's Special Theory of Relativity using previous QuarkNet cosmic ray muon flux studies.
- Studied the effect of zenith angle on cosmic ray muon flux using the QuarkNet Cosmic Ray Muon Detector. Data collected at Irondale High School.
- Proposed and conducted an experiment in cosmic ray muons.

TEACHING EXPERIENCE

University of California, Riverside, Riverside, CA

Mar 2026 – Jun 2026

Graduate Teaching Assistant, Department of Physics and Astronomy

- Teaching assistant for PHYS 2LA: General Physics Laboratory, a course covering the basic principles of classical mechanics for undergraduates.
- Wrote and graded ~ 35 weekly quizzes and laboratory reports.
- Supervised undergraduates in laboratory settings and proctored final laboratory skills assessment.

University of California, Riverside, Riverside, CA

Sep 2025 – Dec 2025

Graduate Teaching Assistant, Department of Physics and Astronomy

- Teaching assistant for PHYS 40A: General Physics, a course introducing classical mechanics to undergraduates.
- Wrote and graded ~ 50 weekly quizzes and laboratory reports.
- Supervised undergraduates in laboratory settings and proctored final laboratory skills assessment.

PUBLICATIONS

Journal Publications

Marx E., Benoit W., Gunny A., et al., Machine-learning Pipeline for Real-time Detection of Gravitational Waves from Compact Binary Coalescences. 2025, PRD, 111

Orlowski-Scherer J., **Venterea R.**, Battaglia N., et al., The Atacama Cosmology Telescope: Millimeter Observations of a Population of Asteroids or: ACTeroids. 2024, ApJ, 964, 2

Venterea R., Orlowski-Scherer J., Næss S., et al., Sub-Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2023, American Astronomical Society Meeting Abstracts, 55, 2:104.12

Venterea R., Ekka U., An Introduction to Quantum Computing. 2022, JURP, 31, 1

Journal Papers in Review

Venterea R.C., Orlowski-Scherer J., Battaglia N., et al., The Atacama Cosmology Telescope: Release of A databaSe of millimeTer ObservatioNs of Asteroids Using acT (ASTRONAUT). 2025, ApJS

Campolongo E., Chou Y.T., Govorkova E., et al., Building Machine Learning Challenges for Anomaly Detection in Science. 2025, Nat. Commun.

Venterea R., Ekka U., An Analysis of Muon Flux from Angle Variation of the QuarkNet Cosmic Ray Detector. 2023, TPT

Conference Papers

(Peer-Reviewed)

Pelosi, D., Barão, F., Bertucci, B., et al., Modeling the Heliospheric Modulation of Cosmic Rays in Light of New Data from AMS-02 in Space. 2025, PoS

Pelosi, D., Barão, F., Bertucci, B., et al., Analysis of Galactic Cosmic Ray Variability and Time-Lagged Relation to Solar Activity. 2025, PoS

Reports

Venterea R. C., Predictions of the Solar Modulation Potential of Cosmic Rays Using Sunspot Number. 2025, U.S. Student Fulbright Program

Doku F., Ekka U., **Venterea R.**, Executive Summary for the Spread of Misinformation and Levels of Censorship. 2021

Muralidhar A., Medvec M., Fujishima B., et al., Preliminary Design Review - Ad Lunam Hopper. 2021, National Aeronautics and Space Administration Lucy Student Pipeline Accelerator and Competency Enabler Mission Concept Academy

Mota A., Sin J., Garcia S., et al., Astronaut-Friendly 3 in 1 Edible Cutlery to Promote Bone Health. 2021, National Aeronautics and Space Administration Lucy Student Pipeline Accelerator and Competency Enabler Proposal Writing and Evaluation Experience Academy

Manuscripts

He Y., **Venterea R.**, Wu X., Food Distribution by Mobile Food Pantries: A Design for Optimized Schedule. 2020

PRESENTATIONS AND INVITED LECTURES

Presentations

Barão F., Bertucci B., Fiandrini E., et al., Analysis of Time Variability of Galactic Cosmic Rays. 2025, International Cosmic Ray Conference 2025

Venterea R., Orlowski-Scherer J., Battaglia, N., Næss S., et al., Sub-Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2024, Emerging Researchers in Exoplanet Science Symposium IX

Venterea R., Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2024, Cornell Undergraduate Research Board Spring Symposium

Venterea R., Galaxies and Asteroids. 2024, 13th Semi-Annual CURBx Research Conference

Marx E., Benoit W., Gunny A., et al., A search for binary mergers in archival LIGO data using aframe, a machine learning detection pipeline. 2024, April Meeting

Benoit W., Marx E., Gunny A., et al., A machine-learning pipeline for real-time detection of gravitational waves from compact binary coalescences. 2024, April Meeting

Venterea R., Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2023, Cornell University Undergraduate Research Poster Forum

Venterea R., Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2023, Cornell Undergraduate Research Board Spring Symposium

Venterea R., Observations of Asteroids with ACT. 2023, University of Rochester 2nd Annual Undergraduate Astronomy Research Seminar

Venterea R., Asteroids and ACT. 2022, Atacama Cosmology Telescope Collaboration Meeting

Venterea R., Looking at Asteroids. 2022, Cornell University Nexus Scholars Program Capstone Presentations

Venterea R., Curvature. 2021, Cornell University DRP Talks

Doku F., Ekka U., **Venterea R.**, Problem C: Submitted a Tweet, Now What? 2021, SIMIODE Challenge Using Differential Equations Modeling

Venterea R., A Brief Introduction to Quantum Field Theory. 2021, Cornell University DRP Talks

Muralidhar A., Medvec M., Fujishima B., et al., Team 21 - Ad Lunam. 2021, National Aeronautics and Space Administration Lucy Student Pipeline Accelerator and Competency Enabler Mission Concept Academy PDR Presentation

Doku F., Ekka U., **Venterea R.**, Using Quantum Computing to Classify Solar Flares. 2021, HackUTD: The VII Seas

Invited Lectures

Venterea R. C., QuarkNet Workshop: My Academic Journey, 2024, QuarkNet, University of Minnesota, Twin Cities

GRANTS

Past Research

Nicola Tomassetti
Fulbright U.S. Student Program
€ 13,800

October 2024 – July 2025

Developing a Web Application for Cosmic-ray and Space Physics Data
Research Assistant

GBT/24B-184 (John Orlowski-Scherer)
National Radio Astronomy Observatory
August 2024 – December 2024

Investigating Anomalous Flux from the Asteroids (511) Davida and (423) Diotima
Research Assistant

HONORS AND AWARDS

Sons of Italy Foundation National Leadership Grant 2026

National Science Foundation Graduate Research Fellowship Program 2026 – *Present*

Hispanic Scholarship Fund Finalist 2022 – 2024, 2025 – 2026

Dean's Distinguished Award 2025 – 2026

JS Enrichment Fund 2024 – 2025

Foundations of Citizen Science 2024

Heising-Simons Foundation Grant 2024

Dean's Distinguished Award, declined	<i>2024 – 2025</i>
Alpha Phi Alpha Fraternity Memorial Scholarship	<i>2020 – 2024</i>
Fulbright U.S. Student Program Scholar	<i>2024 – 2025</i>
Wells Fargo Foundation HSF Scholarship	<i>2023</i>
Discover Scholar at the University of Chicago	<i>2023</i>
Grainger Engineering MERGE Scholar	<i>2023</i>
ASPIRE Illinois Scholar at the University of Illinois Urbana-Champaign	<i>2023</i>
Dean's List of the College of Arts and Sciences for Excellence in Scholarship	<i>2021, 2023</i>
Hispanic Scholarship Fund Scholar	<i>2020 – 2021, 2022 – 2024</i>
Summer Experience Grant	<i>2023</i>
Undergraduate Research Fund	<i>2023</i>
Einhorn Discovery Grant	<i>2023</i>
FOCUS Scholar at the Georgia Institute of Technology	<i>2023</i>
Society of Hispanic Professional Engineers Undergraduate Scholarship	<i>2022 – 2023</i>
Inaugural Session of the Nexus Summer Scholars Program	<i>2022</i>
Meritorious Award for Differential Equations Modeling SIMIODE Challenge	<i>2021</i>
D. E. Shaw Latitude Fellowship	<i>2021</i>
National Name Exchange	<i>2021</i>

COMMUNITY SERVICE

CECI Workshop on Detector Science

Volunteer, June – July 2026

Cleaned, organized posters, and served as presentation timer.

Science x AI Summit

Volunteer, June 2026

Served as receptionist for SAIR participants.

Riverside Unified School District Annual Science & Engineering Fair

Judge, January 2026

Served as a judge for elementary school science fair posters.

UCR Highlander Invitational

Grader, January 2026

Graded Rocks & Minerals Division B of the Science Olympiad tournament.

American Physics Society National Mentoring Community

Mentor, August 2024 – Present

Mentee accepted at MSU, Howard.

Reach the World

November 2024 – April 2025

Described my academic journey abroad to K-12 classrooms.

National Institute of Development Advancement Certified Chapter Leader Program

Facilitator, July 2023

National Academic Quiz Tournaments 2023 High School National Championship Tournament

Full-time scorekeeper, May 2023

Nexus Scholars Program Information Session 2023

Student panelist, October 2022

Zooniverse

Citizen scientist, 2022 – 2023

PROFESSIONAL TRAINING

Fulbright Media Literacy Seminar, Rome, Italy

Mar 2025

Discussed the implications of artificial intelligence in the global geopolitical sphere.

University of Minnesota, Twin Cities, Minneapolis, Minnesota

Jul 2024 – Aug 2024

Zwicky Transient Facility Summer School

University of California, Riverside, Riverside, California

Jul 2024 – Aug 2024

NSF GRFP Application Workshop Series

Coursework

Compute Ontario Advanced Research Computing Training

Jun 2022

Using JupyterLab

Coursework

Compute Ontario Advanced Research Computing Training

Jun 2022

Introduction to Python

Cornell University, Ithaca, New York

May 2022 – Jul 2022

Attended weekly meetings in professional development that emphasized career goals and improving communication and research skills

Coursework

SciNet High Performance Computing Consortium
Databases in Scientific Computing

Jan 2022

Coursework

SciNet High Performance Computing Consortium
Introduction to GPU Programming

Jan 2022

Coursework

SciNet High Performance Computing Consortium
Intro to SciNet, Niagara, and Mist

Dec 2021

Coursework

SciNet High Performance Computing Consortium
Intro to the Linux Shell

Nov 2021

PROFESSIONAL AFFILIATIONS

Association of Latino Professionals For America

2021 – Present

Atacama Cosmology Telescope Collaboration

2021 – Present

Cerro Chajnantor Atacama Telescope Collaboration

2026 – Present

Cosmic Footprint Society

2024 – Present

National Italian American Foundation

2025 – Present

National Society of Hispanic Physicists

2021 – Present

Order Sons and Daughters of Italy in America

2025 – Present

Society for the Advancement of Chicanos and Native Americans in Science

2023 – Present

Society of Hispanic Professional Engineers

2021 – Present

COMPUTER SKILLS

Programming: Python, MATLAB, R, Java, LaTeX, Unix shell, JavaScript, HTML, CSS

Applications: GitHub, Jupyter Notebook, JupyterLab, Google Colab, ArcGIS, JMARS, Visual Studio Code, Google Docs Editors, Microsoft Office, MobaXterm, SigmaPlot, Zotero, Amazon AWS

EXTRACURRICULARS

Future GRADS MentorSHPE	2023 – 2024
Mentoring program for prospective graduate students from underrepresented backgrounds emphasizing networking and writing graduate application essays. Hosted by the Society of Hispanic Professional Engineers.	
SIMIODE Challenge Using Differential Equations Modeling VI	2021
Modeled the spread and censorship of misinformation on social media platforms via differential equations with an epidemiological approach. Presentation won Meritorious Award.	
Directed Reading Program	2021
Learned topics in quantum field theory. Learned topics in differential geometry. Presented on both subject areas.	
HackUTD: The VII Seas	2021
Collaborated in a team of three to implement a quantum computer algorithm to classify solar flare data. Successfully implemented with an accuracy of 98% 24 hours in advance.	
Cornell Society of Physics Students	2020 – 2024
Cornell Math Club	2020 – 2021
Cornell Astronomical Society	2020 – 2024
Cornell Chess Club	2020 – 2024
Competed in the 2023 – 2024 United States Amateur East Championships. Captain of the Cornell C Team.	
Cornell Mathematical Contest in Modeling	2020
Helped develop an algorithm to optimize scheduling for the Food Bank of the Southern Tier.	
Cornell Undergraduate Research Board	2020
Mentoring program for undergraduate students planning research careers emphasizing professional development, presenting research, and interacting with professors.	

LANGUAGES

English: Native Language

Italian: Intermediate in Listening, Speaking, Reading, and Writing

HOBBIES

Learning, stargazing, playing classical music, reading, fishing for small mouth bass, badminton, brewing coffee